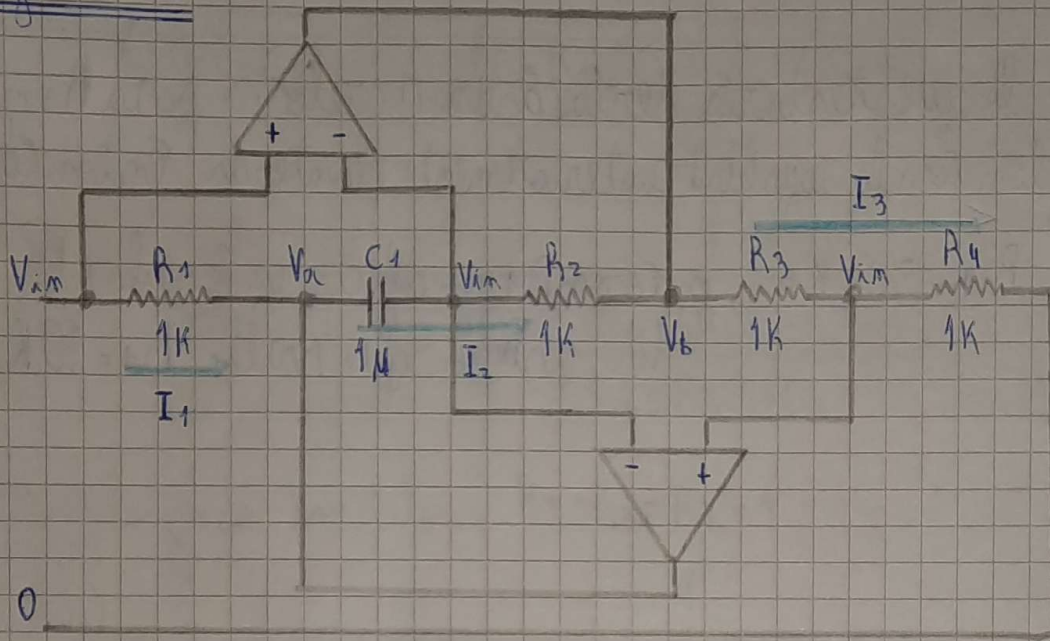


Ejercicio 7



Plantear ecuaciones iniciales:

$$\bullet I_1 = \frac{V_{in} - V_a}{R_1} ; \text{ donde } I_1 = I_{in} \quad (1)$$

$$\bullet I_2 = \frac{V_a - V_{in}}{Z_{C1}} = \frac{V_{in} - V_b}{R_2} \quad (2)$$

$$\bullet I_3 = \frac{V_b - V_{in}}{R_3} = \frac{V_{in} - 0}{R_4} \quad (3)$$

$$\bullet \text{ En (3) despejo a } V_b: \frac{V_b}{R_3} - \frac{V_{in}}{R_3} = \frac{V_{in}}{R_4} \rightarrow \frac{V_b}{R_3} = \frac{V_{in}}{R_3} + \frac{V_{in}}{R_4}$$

$$\frac{V_b}{R_3} = V_{in} \left(\frac{1}{R_3} + \frac{1}{R_4} \right) \rightarrow \frac{V_b}{R_3} = V_{in} \frac{(R_3 + R_4)}{R_3 R_4} \rightarrow V_b = V_{in} \frac{(R_3 + R_4)}{R_4} \quad (4)$$

$$\bullet \text{ Reemplazo (4) en (2): } \frac{V_a - V_{in}}{Z_{C1}} = \frac{V_{in} - V_{in} \frac{(R_3 + R_4)}{R_4}}{R_2}$$

$$\frac{V_a}{Z_{c1}} = \frac{V_{im}}{R_2} + \frac{V_{im}}{Z_{c1}} - V_{im} \frac{(R_3 + R_4)}{R_4 R_2}$$

$$\frac{V_a}{Z_{c1}} = V_{im} \left(\frac{1}{R_2} + \frac{1}{Z_{c1}} - \frac{(R_3 + R_4)}{R_4 R_2} \right)$$

$$\frac{V_a}{Z_{c1}} = V_{im} \frac{(R_4 Z_{c1} + R_4 R_2 - (R_3 + R_4) Z_{c1})}{R_2 R_4 Z_{c1}}$$

$$V_a = V_{im} \frac{(R_4 Z_{c1} + R_4 R_2 - R_3 Z_{c1} - R_4 Z_{c1})}{R_2 R_4}$$

$$V_a = V_{im} \frac{(R_4 R_2 - R_3 Z_{c1})}{R_2 R_4} \quad (5)$$

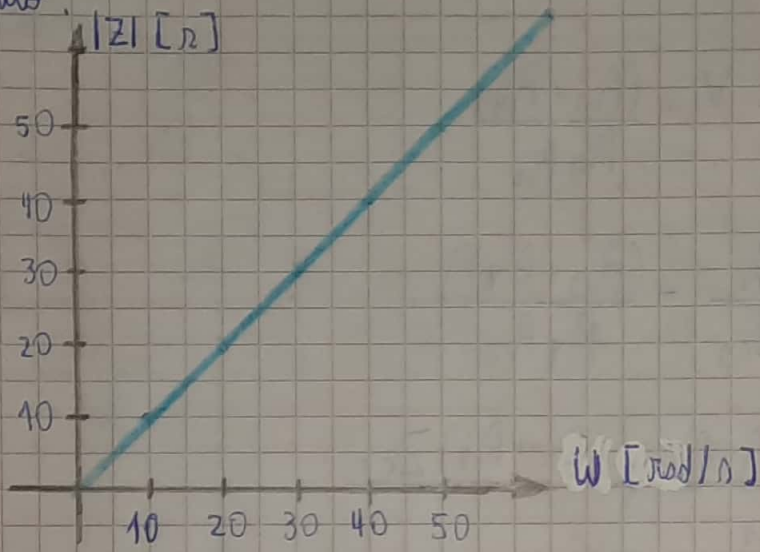
• Reemplazo (5) en (1): $I_{im} = \frac{V_{im}}{R_1} - \frac{V_{im} (R_4 R_2 - R_3 Z_{c1})}{R_2 R_4 R_1}$

$$I_{im} = V_{im} \left(\frac{1}{R_1} - \frac{R_4 R_2 - R_3 Z_{c1}}{R_2 R_4 R_1} \right) \rightarrow I_{im} = V_{im} \left(\frac{R_2 R_4 - R_2 R_4 + R_3 Z_{c1}}{R_1 R_2 R_4} \right)$$

$$Z_{im} = \frac{V_{im}}{I_{im}} = \frac{R_1 R_2 R_4}{R_3 Z_{c1}} = \frac{R_1 R_2 R_4}{R_3} \cdot \frac{1}{Z_{c1}}$$

Reemplazando valores: $Z_{im} = 51 \Omega$

• Módulo



• Fase.

